GRAPH INVARIANTS AND REGULAR MAPS

DAVID HUME AND RYAN LAM

Abstract. We introduce and show that a collection of graph invariants, usually of interest in network design and combinatorialist, are monotone coarse invariants for bounded degree graphs, which is an important condition for geometric group theorists.

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Introduction

Given a graph G, a *graph invariant* is a non-negative integer associated G, which is unchanged under the isomorphisms of graphs. G can also be viewed as a metric space, for which we can discuss 'nice' functions, such as *regular* functions between two graphs X, Y. This paper focuses on connected, bounded degree, and possibly infinite graphs, we survey various graph invariants, which are widely researched in combinatorics, and various bounds are known. We use them to investigate *profiles* of a fixed graph, and show them are monotone by a controlled order under regular maps.

Invariants such as cutwidth and bandwidth are crucial in linear arrangement problems and have been studied extensively. For example, [BKL08, Table 1] contains results from different papers on the complexity of an algorithm for deciding the cutwidth of different types of graphs. There are also been results on upper and lower bounds of bandwidth on some special graphs in [CO86]. Treewidth is studied under [RS83; RS86], [KS93] also contains results on various graph invariants.

On the other hand, the notation of regular maps and degree-bounded graphs is in the interest of geometric group theorists. In particular, given a group G, if H is a subgroup of G, then there is always an immersion of Cayley graphs $\phi : \mathrm{Cay}(H) \to \mathrm{Cay}(G)$ that is regular. Historically, this perspective was popularized in [BST11], for which the graph invariant sep was investigated.

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School of Mathematics, Fry Building, Woodland Road, Bristol, BS8 1UG. Email: ryan.cklam.2020@bristol.ac.uk, URL: ryanlamhk.wordpress.com.

In section 1, we will introduce a collection of graph invariants and notations for our main theorem 1.13. In section 2, we prove some basic properties of different graph invariants and provide a classification of their orders such that the proof of the main theorem is simplified. We will prove our main theorem in section 3, highlighting an important lemma 3.1 for understanding regular maps. Finally, we address any possible research direction and open problems in section 4.

1. Main Results

Given a connected graph G with vertex set VG and edge set EG, we will write $|G|$ as a shorthand for $|VG|$ if it is finite, and $\deg(G)$ to be the maximum degree over all vertices of G. We say G is *degree-bounded* if $\deg(X) < \infty$. We also write $H \leqslant G$ if H is a subgraph of G. There are various ways to measure the "width" on G, including the following.

Definition 1.1. Given a graph G, the **graph geodesic metric** is the function $d_G : VG \times VG \to \mathbb{R}$ which sends two vertices (v, w) to the length of the shortest path connecting v and w.

Note that this makes (VG, d_G) a metric space for any graph G.

Other than the graph geodesic metric, there are also some other measurements of "width". In the next few definitions, set G to be a finite graph.

Definition 1.2. A **cutset** of G is some $S \subseteq VG$ such that $G \setminus S$ has no connected component with greater than $\frac{|G|}{2}$ vertices.

The **cutsize** of G, denoted $\mathrm{cut}(G)$, is the minimal cardinality of a cutset of G.

Definition 1.3. A **numbering** is a bijective function $VG \to \{1, 2 \cdots |VG|\}$.

In this case, the **cutwidth** of G, denoted by $\mathrm{cw}(G)$, is defined as

$$\mathrm{cw}(G) := \min_f \max_{1 \leqslant k \leqslant |VG|} \#\{(x, y) \in EG : f(x) < k \leqslant f(y)\}.$$

The **bandwidth** of G, denoted by $\mathrm{bw}(G)$, is defined as

$$\mathrm{bw}(G) := \min_f \max_{(x,y) \in EG} |f(x) - f(y)|,$$

where f ranges over all possible numberings of the graph G.

Definition 1.4. The **modified cutwidth** of G, denoted by $\mathrm{mcw}(G)$, is defined as

$$\mathrm{mcw}(G) := \min_f \max_{1 \leqslant k \leqslant |VG|} \#\{(x, y) \in EG : f(x) < k < f(y)\}.$$

where f ranges over all possible numberings of the graph G.

Definition 1.5. The **topological bandwidth** of G, denoted by $\mathrm{tbw}(G)$, is defined as

$$\mathrm{tbw}(G) := \min\{\mathrm{bw}(\widetilde{G}) : \widetilde{G} \text{ is obtained by finitely many edge subdivisions from } G\}$$

where f ranges over all possible numberings of the graph G.

Definition 1.6. Let T be a finite tree, and let $(X_i : i \in I)$ be a family of subsets of VG such that $VT = I$. An element $X_i \subseteq VG$ in the family is often referred to as a **node vertex** or a **pot**.

A **tree decomposition of** G is a pair (T, X_i) (or (X_i, T)) as above such that the following properties is satisfied:

(a) $\bigcup_{i \in I} X_i = V(G)$.
(b) If $(u, v) \in EG$, then there is $i \in VT$ such that $u \in X_i$ and $v \in X_i$.
(c) For $i, j, k \in I$, if j lies on the path of T from i to k then $X_i \cap X_k \subseteq X_j$.

The **width** of the tree-decomposition, denoted by $w(X_i, T)$ is

$$w(X_i, T) := \max\left(|X_i| - 1 : i \in I\right).$$

The **treewidth** of G, denoted by $\mathrm{tw}(G)$ is the minimum $w \geqslant 0$ such that G has a tree-decomposition of width not more than w.

Definition 1.7. The **pathwidth** has almost exactly the same definition of treewidth, except for tree decomposition we have **path decomposition**, where the decomposition (T, X_i) requires T to be a path instead of a tree.

Definition 1.8 (Informal)**.** The **search number** of a graph G is introduced at the introduction of [Meg+81], which involves the number of 'searchers' needed to 'clear' a contaminated graph.

All of these invariants are defined for finite graphs. To study these invariants on infinite graphs we introduce the **profile** function.

Definition 1.9 (Profiles)**.** Let X be a graph. We define the M**-profile** as a function of type $\mathbb{N} \to \mathbb{N}$:

$$M_X(n) = \max\{M(\Gamma) \mid \Gamma \subset X, |V\Gamma| \leqslant n\}$$

where $M \in \{\mathrm{tw}, \mathrm{cw}, \mathrm{tbw}, \mathrm{s}, \mathrm{pw}, \mathrm{bw}\}$.

If $M = \mathrm{cut}$ this is referred to as the **separation profile** instead of the *cut-profile*, and we write $\mathrm{sep}_X(n)$ instead of $\mathrm{cut}_X(n)$.

We only consider monotonicity up to the big-O order relation, that is:

Definition 1.10. Given two functions $f, g : \mathbb{N} \to \mathbb{R}$, we write $f \lesssim g$ if there exists a constant $C > 0$ such that for all $n \in \mathbb{N}$,

$$f(n) \leqslant Cg(Cn) + C$$

and we write $f \simeq g$ if $f \lesssim g$ and $g \lesssim f$.

From the perspective of geometric group theorists, we want to distinguish between graphs with different degrees, so in the relation above, we allow the constant C to depend on $\deg(G)$. Our first result is towards categorizing some extensively studied graph invariants from this perspective.

Theorem 1.11. For all degree-bounded graphs G:

$$\mathrm{sep}_G \lesssim \mathrm{tw}_G \lesssim \mathrm{cw}_G \simeq \mathrm{tbw}_G \simeq \mathrm{s}_G \simeq \mathrm{pw}_G \lesssim \mathrm{bw}_G.$$

The proof will be largely from collecting existing literature, which is presented in section 2.

Another important perspective we will consider is functions between graphs. With the geodesic metric, it can be viewed as functions between metric spaces. There is one particular type of map that is of interest.

Definition 1.12. Let X, Y be metric spaces, a map $\phi : X \to Y$ is κ**-regular** if there is $\kappa \geqslant 1$ such that

(a) $d_Y(\phi(u), \phi(v)) \leqslant \kappa\left(1 + d_X\left(u, v\right)\right)$ for all $u, v \in X$, and
(b) $\left|\phi^{-1}(y)\right| \leqslant \kappa$ for all $y \in Y$.

ϕ is **regular** if it is κ-regular for some $\kappa \geqslant 1$.

Under regular maps of bounded degree graphs $\phi : X \to Y$, [BST11, Lemma 1.3] has proved that $\mathrm{sep}_X \lesssim \mathrm{sep}_Y$, with the constant depending on $\kappa, \deg(X), \deg(Y)$. We generalize their result as follows:

Theorem 1.13 (= Theorem 3.6, 3.9, 3.12)**.** Let X, Y be bounded degree graphs. If there is a regular map $\phi : X \to Y$ then,

$$M_X \lesssim M_Y$$

where $M \in \{\text{sep}, \text{tw}, \text{cw}, \text{tbw}, \text{s}, \text{pw}, \text{bw}\}$.

Note that due to theorem 1.11 and the result of [BST11] above, it suffices to show the statement holds for $M \in \{\text{tw}, \text{cw}, \text{bw}\}$. This is done in section 3.

2. Graph Invariants

Now, we take a close look at the various graph invariants introduced in section 1. Firstly, we show that when fixing a finite graph G, most of the invariants are smaller under a subgraph $H \leqslant G$, with the exception of cut. Next, we present known bounds relating different invariants, and theorem 1.11 will immediately follow.

Note that all the invariants are non-negative, hence there must be a minimum-achieving numbering or tree decomposition for any finite graph G. This would be very useful in proving inequalities while manipulating the minimums and maximums in the definition. Clearly, if G is a path, then all the invariant introduced above is 1, with the exception of mcw being 0.

Proposition 2.1. Given a finite graph G, if $H \leqslant G$, then $M(H) \leqslant M(G)$ for $M \in \{\text{bw}, \text{cw}, \text{tw}\}$.

Proof. We first handle $M = \text{bw}, \text{cw}$. Note that for any graph G, and any numbering f given to G, it restricts an injective function $f\mid_H : VH \to \{1, 2 \cdots |VG|\}$. *A priori* $f\mid_H$ is not a numbering, but we can order vertices in $H, h_1, h_2, \cdots h_m$ such that

$$f(h_1) < f(h_2) \cdots f(h_m),$$

then defining a new function $\widehat{f}$ with $\widehat{f}(h_i) := i$ makes $\widehat{f}$ a numbering of H. Clearly we have for all $u, v \in VH$, $\widehat{f}(u) < \widehat{f}(v)$ implies $f(u) \leqslant f(v)$. Hence, one can check that we have

$$\#\{(u', v') \in EH : \widehat{f}(u') < t \leqslant \widehat{f}(v')\} \leqslant \#\{(u, v) \in EG : f(u) < f(\widehat{f}^{-1}(t)) \leqslant f(v)\} \tag{1}$$

for all $1 \leqslant t \leqslant |VH|$. Also, we have

$$|\widehat{f}(x) - \widehat{f}(y)| \leqslant |f(x) - f(y)| \quad \text{for all } x, y \in VH \tag{2}$$

by the definition of $\widehat{f}$. Hence we can put maximum over both sides in equation (1) and (2). Now we pick the cutwidth (resp. bandwidth) minimizing numbering f, then we have

$$\max_t \#\{(u', v') \in EH : \widehat{f}(u') < t \leqslant \widehat{f}(v')\} \leqslant \min_f \max_k \#\{(u, v) \in EG : f(u) < k \leqslant f(v)\} \tag{3}$$

$$\max_{(x,y)\in EH} |\widehat{f}(x) - \widehat{f}(y)| \leqslant \min_f \max_{(x,y)\in EG} |f(x) - f(y)| \tag{4}$$

hence $\text{cw}(H) \leqslant \text{cw}(G)$ and $\text{bw}(H) \leqslant \text{bw}(G)$ as required.

For $M = \text{tw}$, it is easy to see that if (T, X_i) is a tree decomposition of G, then $(T, X_i \cap VH)$ is a tree decomposition of H: this is simply due to (T, X_i) being a tree decomposition. In particular, to see the third condition, note that for all i, j, k, we have $X_i \cap X_k \subseteq X_j$ implies $(X_i \cap VH) \cap (X_k \cap VH) \subseteq X_j \cap VH$. □

This is not true for cut, see the example in Proposition 2.9.

Below is a collection of bounds from different literatures:

Theorem 2.2 (Theorem 2.2, [MPS83])**.** If $\deg(X) = 3$, then $\text{mcw}(X) + 1 = \text{tbw}(G)$.

Theorem 2.3 (Corollary 2.1, [MPS83])**.** For all finite graphs G, $\text{tbw}(G) \leqslant \text{cw}(G)$.

Theorem 2.4 (Corolalry 2.2, [MPS83])**.** For all finite graphs G, $\mathrm{s}(G) \leqslant \mathrm{tbw}(G) + 1$.

Theorem 2.5 (Lemma 1, Lemma 3, [KS93])**.** For all finite graphs G, $\mathrm{pw}(G) \leqslant \mathrm{cw}(G) \leqslant \mathrm{pw}(G) \cdot \deg(G)$.

Theorem 2.6 (Lemma 3.1, Theorem 3.2, [MS89])**.** For all graphs G, $\mathrm{s}(G) \leqslant \mathrm{cw}(G) \leqslant \left\lfloor \frac{\deg G}{2} \right\rfloor \cdot (s(G) - 1) + 1$.

In particular, [BST11] mentioned a lemma, which we now provide the proof in detail.

Proposition 2.7 (Lemma 2.3, [BST11])**.** For a graph X, $\mathrm{sep}_X(n) \lesssim \mathrm{tw}_X(n)$.

Before proving this theorem, we need a Lemma on the effect of removing node vertices on a tree decomposition of a graph.

Lemma 2.8. Let (T, X_i) be a tree decomposition of a finite graph G. For $i \in VT$, let C_1, C_2 be two distinct components of $T \setminus \{i\}$, then the two induced subgraphs

$$\bigcup_{t \in C_1} X_t \setminus X_i, \quad \bigcup_{t \in C_2} X_t \setminus X_i$$

of G are disjoint from each other.

Proof. First we show there are no common vertex, for if there is, there will be node vertices $t_1 \in C_1, t_2 \in C_2$ such that $v \in VG$ with $v \in X_{t_1} \setminus X_i$ and $v \in X_{t_2} \setminus X_i$. Note that i is in the unique path connecting t_1 and t_2 since C_1, C_2 are distinct connected components of $T \setminus \{i\}$, hence by condition 1.6c we must have $v \in X_i$, which is a contradiction.

Now we show there are no edge $(v, w) \in EG$ connecting $v \in\in X_{t_1} \setminus X_i$ and $w \in X_{t_2} \setminus X_i$. For if there is, then by Condition 1.6b there is node vertices X_{t_3} such that $v \in X_{t_3} \setminus X_i$, $w \in X_{t_3} \setminus X_i$, now if $t_3 \in C_1$, then the unique path connecting t_2 to t_3 in T passes through i, which by condition 1.6c again means $w \in X_i$, contradiction. Similarly, t_3 cannot be in C_2 by considering v. Hence t_3 is in another connected component of $X \setminus \{i\}$, but that also would imply the unique path of t_1 to t_3 contains i, that again imply $v \in X_i$, which is a contradiction. □

Proof of Proposition 2.7. It suffices to show that for any finite graph G, we have $\mathrm{cut}(G) \leqslant \mathrm{tw}(G) + 1$. To do this, fix G and let (T, X_i) be a tree decomposition of G with treewidth $\mathrm{tw}(G)$.

We claim that there is $\ell \in VT$ such that for all connected component L of $T \setminus \{\ell\}$, we have $\#\left(\bigcup_{t \in L} X_t \setminus X_\ell\right) \leqslant |G|/2$. We do this by an 'pick and terminate' technique. Consider the following steps:

- Set $i \in VT$.
  - If $\max_C \#\left(\bigcup_{t \in C} X_t \setminus X_i\right) \leqslant |G|/2$ where C ranges through all connected component of $T \setminus \{i\}$, then we are done as $\ell = i$ is a required vertex.
  - Or else, we have $\max_C \#\left(\bigcup_{t \in C} X_t \setminus X_i\right) > |G|/2$. By Lemma 2.8, there is a unique component T_1 of $T \setminus \{i\}$ such that $\left(\bigcup_{t \in T_1} X_t \setminus X_i\right) > |G|/2$, note that there is a unique vertex $j \in VT_1$ (distinct to i) which neighbours i.
- Now we replace i to j and repeat the procedure, that is, let $\widetilde{C}$ be the connected components of $T \setminus \{j\}$,
  - If $\max_{\widetilde{C}} \#\left(\bigcup_{t \in \widetilde{C}} X_t \setminus X_j\right) \leqslant |G|/2$ then $\ell = j$ and we are done.
  - Otherwise, there is unique component T_2 of $T \setminus \{j\}$ such that $\#\left(\bigcup_{t \in T_2} X_t \setminus X_j\right) > |G|/2$ and a unique vertex $k \in V\widetilde{T}$ connecting j.
- Now we replace $j = k$ and investigate the connected components of $T \setminus \{k\}$, and investigate as above.

To show the existence of ℓ, we must show the steps above can not continue indefinitely. First note that we assumed $\#\left(\bigcup_{t\in T_1} X_t \setminus X_i\right) > |G|/2$ and $\#\left(\bigcup_{t\in \widetilde{T}_2} X_t \setminus X_j\right) > |G|/2$, there must be an common element v in the two sets above. If $i = k$, then $T_1 \cup T_2 = T$ and the union is disjoint. Now, consider the common vertex v, again $v \in X_{t_1} \setminus X_i$ and $v \in X_{t_2} \setminus X_j$ for some $t_1 \in T_1$, $t_2 \in T_2$, the unique path connecting t_1, t_2 must contain both i and j. Hence we have $v \in X_i \cap X_j$, which is a contradiction, so it follows that $i \neq k$. Now in the above steps we have picked out vertex $(i, j, k \cdots)$, with $i \neq k$, we are sure that each step we are picking distinct vertex along a path (heuristically, each vertex is 'farther away' from i). Hence, since T is finite, the process stops, and hence such ℓ must exist.

With X_ℓ removed from G, by Lemma 2.8, X_ℓ indeed a cutset for G, since X_ℓ has cardinality at most $\mathrm{tw}(G) + 1$, the inequality is proven as required. □

Note that the collection of Theorem 2.2 to Proposition 2.7 combined implies Theorem 1.11 by manipulating inequalities.

There is a second part to [BST11, Lemma 2.3], but the author believes it is incorrect. Below is the negation of it.

Proposition 2.9. There is a graph G such that, for some c, for all k such that $\mathrm{tw}(G) \geqslant k$, we have $\mathrm{cw}(G) < c$.

Proof. We take $c = 2$, and consider a graph G which contains two copies of the complete graph K_n, with an extra vertex connecting a vertex from each copy:

$$G: \quad K_n \bullet \text{——} \bullet \text{——} \bullet K_n$$

By the monotonicity of treewidth (Proposition 2.1), G has a treewidth of at least $n - 1$. But since $|G| = 2n + 1$, the middle vertex alone is a cutset for G, hence $\mathrm{cut}(G) = 1$. □

Note that although the counterexample raised is not degree-bounded, but the argument can be generalized with anything instead of K_n with increasing treewidth. The above is also an example of cut does not exhibit monotonicity under subgraphs (see Proposition 2.1), because $\mathrm{cut}(K_n) = n/2$ as removing i vertices of K_n yields K_{n-i}.

3. Profiles under Regular Maps

The following section is devoted to proving one of our main results Theorem 1.13. As discussed, it suffices to prove the monotonicity result on bw, cw and tw. All graphs in this section will be connected.

Note that the degree-bounded condition on X and Y is necessary because, for any graph X, there is always a 1-regular map from X to the infinite star graph S_∞. To prove the profile satisfies the monotonicity relation by $\lesssim$, we need to show that for all finite subgraph $\Gamma \leqslant X$, there is a finite subgraph $\Gamma' \leqslant Y$, such that the invariant of Γ' is bounded by a constant C multiplied by the invariant of Γ. Note that this constant C can depend on $\kappa, \deg(X)$, and $\deg(Y)$.

In section 3.1, we provide a few important lemmas. In particular, an algorithm for producing Γ' from Γ is presented in detail. It will be reused and modified in other lemmas, and the proof of theorem 1.13. We will split the job of proofing theorem 1.13 into three subsections 3.2 to 3.4, handling the invariants one by one.

3.1. Image of Regular Maps on a Graph.

As mentioned above, the following lemma provides a way to 'descend' the subject of studying profiles to studying finite subgraphs of it. In particular, we will reuse the notation in the algorithm presented in this lemma.

Lemma 3.1. If $\phi : X \to Y$ is a κ-regular map, then for any subgraph $\Gamma \leqslant X$, there is a subgraph $\Gamma' \leqslant Y$ such that:

(a) $\phi(v) \in V\Gamma'$ for all $v \in V\Gamma$, and
(b) $\frac{1}{\kappa}|\Gamma| \leqslant |\Gamma'| \leqslant \kappa \deg(X)|\Gamma|$.

Proof. Consider the following algorithm given Γ, ϕ:

(1) First partition $V\Gamma$ into sets by the equivalence relation

$$x \sim y \qquad \text{if and only if} \qquad \phi(x) = \phi(y)$$

and denote the partition set $\Phi_v := \phi^{-1}(\phi(v))$
(2) Define the following set

$$\mathcal{S} := \{\{\Phi_v, \Phi_{v'}\} : [v] \in V\Gamma/\sim, [v'] \in V\Gamma/\sim, [v] \neq [v'], \text{ there is } w \in \Phi_v, w' \in \Phi_{v'} \text{ such that } (w, w') \in E\Gamma\}$$

then choose an element in $\mathcal{S}$ and exactly one of any respective edge $ww' \in E\Gamma$.

By κ-regularity of ϕ and substituting $d_X(w, w') = 1$ in definition 1.12a, find a path connecting $\phi(w)$ and $\phi(w')$ in Y, with length not more than 2κ. Include all the vertices and edges of the path, including the vertices $\phi(w)$ and $\phi(w')$, into $V\Gamma'$ and $E\Gamma'$ respectively.

(3) Repeat step 2 by picking a new element in $\mathcal{S}$. The algorithm stops when all elements of $\mathcal{S}$ have been exhausted.

Remark. The set $\mathcal{S}$ is acting as the "set of edges" of Γ in the construction, but we need this to prevent overcounting. When ϕ is injective, then $\mathcal{S} = E\Gamma$, hence it is obvious that $|S| \leqslant |E\Gamma|$.

Using this algorithm, since G is connected, going through all the edges must go through all the vertices, hence the condition 3.1a is satisfied.

Now we verify the inequality 3.1b from left to right, the first inequality follows from Definition 1.12b as the upper bound of the cardinality of preimage of ϕ. For the second inequality, by the handshaking lemma:

$$|E\Gamma| = \frac{\sum_{v \in V\Gamma} \deg(v)}{2} \leqslant \frac{\deg(X) \cdot |\Gamma|}{2},$$

In step 2, the addition of vertices happens at most $|\mathcal{S}|$ times, and $|\mathcal{S}| \leqslant |E\Gamma| \leqslant \deg(X) \cdot |\Gamma|/2$. Each time the step is called, there were at most 2κ vertices added. Since this is the only step where vertices are added to $V\Gamma'$, we have

$$|\Gamma'| \leqslant \frac{\deg(X) \cdot |\Gamma|}{2} \cdot 2\kappa = \kappa \deg(X)|\Gamma|$$

as required. □

Lemma 3.2. Let G be a graph of bounded degree and fix $v \in VG$ and $k \in \mathbb{N}$. Then there are at most $k \cdot \deg(G)^k$ paths of length $\leqslant k$ with distinct start and end points containing v as a non-endpoint vertex.

Proof. We argue this by 'building' paths out of v. Let d denote the graph geodesic metric, and note that in G, the number of vertices w such that $d(v, w) \leqslant k$ is bounded by $\deg(G)^k$. More explicitly, if there are ℓ vertices w such that $d(v, w) = i$, then there are at most $(\deg(G) - 1)\ell$ vertices w with $d(v, w) = i + 1$, it follows that there are at most $\deg(G)^k$ such vertices.

Now we count the number of paths P of length $l > 2$ with endpoints x and y. Suppose v is vertex r in the path, i.e. $d(x, v) = r$. From v through to x, we have to choose r vertices and have at most $\deg(G) - 1$ to choose from at each step. Similarly, we have $l - r - 1$ vertices to choose to get to y. Thus we have $(\deg(G) - 1)^r \times (\deg(G) - 1)^{l-r-1} = (\deg(G) - 1)^{l-1}$ possible paths

with r fixed. There are $l-2$ choices for r. Thus, we have $(l-2)(\deg(G)-1)^{l-1}$ possible paths of length l passing through v. Thus we have,

$$\sum_{l=2}^{k}(l-2)(\deg(G)-1)^{l-1} \leqslant k\deg(G)^k,$$

hence there are at most $k\deg(G)^k$ paths of length $\leqslant k$ passing through v with distinct endpoints, as required. □

Definition 3.3. A map of vertices between graphs $\phi : VX \to VY$ is a **simplicial map** if for all $(x, x') \in EX$, *at least one* of the following holds:

(a) $\phi(x) = \phi(x')$, or
(b) $(\phi(x), \phi(x')) \in EY$.

Proposition 3.4. Given any (possibly infinite) graphs X, Y, if there is a κ-regular map $\phi : X \to Y$, then for any finite subgraph $\Gamma \leqslant X$, there is a finite subgraph Γ' and functions f, g such that the following diagram commutes:

$$\begin{array}{ccc} \Gamma & \xrightarrow{\phi} & \Gamma' \\ & \searrow^{f} & \uparrow{\scriptstyle g} \\ & & \gamma \end{array}$$

, where γ is a subdivision of Γ (hence f is the canonical injection), and g is a simplicial map.

Proof. Use Lemma 3.1 and the algorithm presented, find $\Gamma' \leqslant Y$. Now, using the notation from the algorithm, given $\phi : \Gamma \to \Gamma'$, we construct the subdivision γ of Γ and the map $g : \gamma \to \Gamma'$ as follows:

(1) Start by considering an element $\{\Phi_v, \Phi_{v'}\}$ in $\mathcal{S}$.
(2) By κ-regularity and the algorithm, there is a path
$$\mathcal{P}_{(v,v')} := (\phi(v) =: w_0, w_1, w_2, \cdots, w_m := \phi(v'))$$
(3) By definition of an element in $\mathcal{S}$, there is an edge $E := (w, w') \in E\Gamma$ such that $w \in \Phi_v, w' \in \Phi_{v'}$. Subdivide this edge E to a path $\widetilde{E}$ as follows:
$$\widetilde{E} := (w, \widetilde{w_1}, \widetilde{w_2}, \cdots \widetilde{w_{m-1}}, w')$$
where $\widetilde{w_1}, \widetilde{w_2}, \cdots \widetilde{w_{m-1}}$ are added vertices, and define g such that
$$g(v) = \begin{cases} \phi(v) & \text{if } v \in \Gamma \\ w_j & \text{if } v = \widetilde{w_j} \end{cases}. \tag{5}$$
(4) Repeat step 3 until every edge $E := (w, w') \in E\Gamma$ such that $w \in \Phi_v, w' \in \Phi_{v'}$ is chosen.
(5) Repeat step 1 until every element of $\mathcal{S}$ is chosen.

Remark. Note that an edge $(v, v') \in E\Gamma$ is picked in step 3 if and only if $\phi(v) \neq \phi(v')$ (since all the edges are picked, and it went through all the elements of $\mathcal{S}$.), equivalently, there is a path $\mathcal{P}_{(a,b)}$ in step 3 such that (v, v') is contained in it.

Now f is the (canonical) injection to γ, where $f(v) = v$ for all $v \in V\Gamma$, hence the diagram commutes by (5).

Now it suffices to show that g is simplicial. For any edge $(v, v') \in E\gamma$, without loss of generality assume it is picked by the algorithm (since otherwise condition 3.3a is automatically satisfied). Since it is picked out, every edge is sent to another edge by the construction of g. Hence condition 3.3b is satisfied for those edges, and g is simplicial as required. □

3.2. **Monotonicity of Bandwidth.**

Proposition 3.5. Given κ-regular $\phi : X \to Y$ and finite Γ such that $\Gamma \leqslant X$, there is finite $\Gamma' \leqslant Y$ such that for every numbering g on Γ', and every $(v, v') \in E\Gamma$, there is numbering f on Γ and $(w, w') \in E\Gamma'$ such that

(a) $d_Y(\phi(v), w)) \leqslant 2\kappa$, and
(b) $|f(v) - f(v')| \leqslant 2\kappa^2 |g(w) - g(w')|$.

Proof. Choose Γ' using Lemma 3.1. We can always choose f such that

$$\text{for all} u, v \in \Gamma : \text{ if } f(u) \geqslant f(v), \text{ then } g(\phi(u)) \geqslant g(\phi(v)). \tag{6}$$

This always exists, because we can always number the vertices in Γ by going preimages in order using g. Without loss of generality, we can assume $f(v) < f(v')$. Hence, we write $f(v) = t, f(v') = t + \ell$ for some t, ℓ. We divide into two cases:

First Case. $|f(v) - f(v')| \leqslant 2\kappa^2$. Take $w = \phi(v)$ and w' to be any edge connected to w (there is at least one since Γ' is connected). Then 3.5a is trivial as the left hand side is 0, 3.5b is obviously true since $|g(w) - g(w')| > 1$ unconditionally.

Second Case. $|f(v) - f(v')| > 2\kappa^2$. As a shorthand, we will denote $f^{-1}(t) = v_t$ for any t. Now consider the vertices below:

$$v =: v_t, v_{t+1}, \cdots v_{t+\ell} := v'$$

in Γ. By the numbering chosen with property (6), we have

$$g(\phi(v_t)) \leqslant g(\phi(v_{t+1})), \cdots \leqslant g(\phi(v_{t+\ell})). \tag{7}$$

Moreover, Definition 1.12b also ensures ϕ is at most κ-to-1, and noting there are $\ell + 1$ vertices in consideration, we must have

$$\#\{g(\phi(v_t)), g(\phi(v_{t+1})), \cdots, g(\phi(v_{t+\ell}))\} \geqslant \frac{\ell+1}{\kappa} > \frac{\ell}{\kappa}. \tag{8}$$

Combining (7) and (8) we have

$$|g(\phi(v_{t+\ell})) - g(\phi(v_t))| > \frac{\ell}{\kappa}. \tag{9}$$

Now given any $v, v' \in V\Gamma$, take the path

$$(\phi(v) =: w_0, w_1, w_2, \cdots, w_m := \phi(v'))$$

in Γ' that connects $\phi(v)$ and $\phi(v')$, where $m \leqslant 2\kappa$ by the construction in Lemma 3.1. Note that there must exist $1 \leqslant i < m$ such that $|g(w_{i+1}) - g(w_i)| > \dfrac{\ell}{2\kappa^2}$, or else if for all $1 \leqslant i < m$ we have $|g(w_{i+1}) - g(w_i)| \leqslant \dfrac{\ell}{2\kappa^2}$:

$$|g(\phi(v_{t+\ell})) - g(\phi(v_t))| = |g(w_m) - g(w_0)| \leqslant \sum_{i=1}^{m} |g(w_i) - g(w_{i-1})| \leqslant \frac{\ell}{2\kappa^2} \cdot m \leqslant \frac{\ell}{2\kappa^2} \cdot 2\kappa = \frac{\ell}{\kappa}$$

by triangle inequality, which contradicts (9). Hence take $w = w_i, w' = w_{i+1}$, we have $|g(w) - g(w')| \geqslant \dfrac{\ell}{2\kappa^2}$ and hence 3.5b reads $|f(v) - f(v')| \leqslant \ell$, which is true.

□

Theorem 3.6. Let X, Y be bounded degree graphs. If there is a regular map $\phi : VX \to VY$, then,

$$\mathrm{bw}_X \lesssim \mathrm{bw}_Y.$$

Proof. Consider any $\Gamma \leqslant X$ with $|\Gamma| \leqslant r$, then for any $(v, v') \in E\Gamma$, and any numbering g for Γ', use Proposition 3.5b to find numbering f in Γ, $(w, w') \in E\Gamma'$ such that $|f(v) - f(v')| \leqslant 2\kappa^2 |g(w) - g(w')|$, note the left hand side must attain maximum for some $(v, v') \in E\Gamma$, so it follows that

$$\max_{(v,v')\in E\Gamma} |f(v) - f(v')| \leqslant 2\kappa^2 |g(w) - g(w')|$$

for some $(w, w') \in E\Gamma'$, which means:

$$\max_{(v,v')\in E\Gamma} |f(v) - f(v')| \leqslant \max_{(w,w')\in E\Gamma'} 2\kappa^2 |g(w) - g(w')|$$

for all numbering g, picking some numbering f. Using similar reasoning, there must be some g which attains the minimum, so there is numbering f which

$$\max_{(v,v')\in E\Gamma} |f(v) - f(v')| \leqslant \min_{g} \max_{(w,w')\in E\Gamma'} 2\kappa^2 |g(w) - g(w')|$$

, hence

$$\min_{f} \max_{(v,v')\in E\Gamma} |f(v) - f(v')| \leqslant \min_{g} \max_{(w,w')\in E\Gamma'} 2\kappa^2 |g(w) - g(w')|$$

$$\mathrm{bw}(\Gamma) \leqslant 2\kappa^2 \mathrm{bw}(\Gamma') \tag{10}$$

$$\mathrm{bw}(\Gamma) \leqslant \max\{2\kappa^2 \mathrm{bw}(\Gamma') : \Gamma' \leqslant Y, |\Gamma'| \leqslant 2\kappa \deg(X)|\Gamma|\} \qquad \text{by Lemma 3.1.}$$

$$\mathrm{bw}(\Gamma) \leqslant 2\kappa^2 \mathrm{bw}_Y(2\kappa \deg(X)|\Gamma|) \tag{11}$$

now for arbitrary Γ such that $|\Gamma| \leqslant r$, (11) holds, it follows that $\mathrm{bw}_X(r) \lesssim \mathrm{bw}_Y(r)$ as required. □

Corollary 3.7. For any finite graphs X, Y, if there is a κ-regular map $\phi : X \to Y$, then

$$\mathrm{bw}(X) \leqslant 2\kappa^2 \mathrm{bw}(Y). \tag{12}$$

3.3. Monotonicity of Cutwidth.

Proposition 3.8. For every numbering g on Γ', and any t such that $1 \leqslant t < |\Gamma|$, there is a numbering f on Γ, constant C depending on $\kappa, \deg X$ and $\deg Y$ such that

$$\#\{v, v' \in E\Gamma : f(v) \leqslant t < f'(v)\} \leqslant C \cdot \#\{w, w' \in E\Gamma' : g(w) \leqslant g(\phi(f^{-1}(t))) < g(w')\} \tag{13}$$

Proof. To simplify notation, we will denote the set on the left hand side as F_t, and right hand side as G_t. We will actually show there is a constant C such that $\#F_t - \kappa \deg(X) \leqslant C \cdot G_t$ for all t. To show this, we define

$$\widetilde{F}_t := F_t \setminus \{(v, v') \in E\Gamma : v \in \Phi_{f^{-1}(t)} \text{ or } v' \in \Phi_{f^{-1}(t)}\}.$$

Note that since t is fixed, by κ-regularity condition 1.12b there are at most κ vertices in $\Phi_{f^{-1}(t)}$, each vertices contains at most $\deg(X)$ edges in F_t, hence there are at most $\kappa \deg(X)$ edges where $v \in \Phi_{f^{-1}(t)}$ or $v' \in \Phi_{f^{-1}(t)}$. It follows that $\#\widetilde{F}_t \geqslant \#F_t - \kappa \deg(x)$.

Now, we prove $\#\widetilde{F}_t \leqslant C \cdot \#G_t$. For each edge $(w, w') \in G_t$, we need to correspond it will a collection of edges in Γ, such that

- For each edge in G_t, there are at most C edges from $\widetilde{F}_t$ in each correspondence, and
- All the edges in $\widetilde{F}_t$ have been corresponded by at least one edge in G_t.

We first present the correspondence below as two cases: Given $(w, w') \in G_t$:

Case 1. If $g(w) = g(\phi(f^{-1}(t)))$. Then we have $\phi(f^{-1}(t)) = w$. Hence by κ-regularity condition 1.12b, there are at most $\kappa \deg(X)$ edges $(v, v') \in E\Gamma$ such that $\phi(v) = w$. Correspond all such edges to (w, w').

Case 2. If $g(w) < g(\phi(f^{-1}(t)))$, if further $w = \phi(v)$ for some $v \in E\Gamma$, then correspond all (at most) $\kappa \deg(X)$ edges as in Case 1. Otherwise, by the construction of Γ', w is an interior vertex contained in a path with length less than 2κ and endpoint $\phi(v), \phi(v')$ for an edge $(v, v') \in E\Gamma$. By Lemma 3.2 there is at most $2\kappa \deg(Y)^{2\kappa}$ such distinct edge. Correspond those edges to (w, w') with the extra requirement that the path $\mathcal{P}_{(v,v')}$ must contain both w and w'.

Now we show the correspondence above indeed satisfies the requirements listed above. Clearly we are corresponding less than $C := \max\{\kappa \deg(X), 2\kappa \deg(Y)^{2\kappa}\}$ edges in Γ for a single edge $(w, w') \in G_t$, so the first condition is satisfied.

To verify the second requirement. Given any $(v, v') \in \widetilde{F}_t$, we need to find an edge in G_t such that the correspondence contains (v, v'). First, notice that there are no edges in $\widetilde{F}_t$ with $\phi(v) = \phi(v')$. Indeed, suppose the contrary that $\phi(v) = \phi(v')$. Since $(v, v') \in F_t$, it follows that $f(v) \leqslant t < f(v')$, and by the property of the numbering (6) we have $g(\phi(v)) \leqslant g(\phi(f^{-1}(t))) \leqslant g(\phi(v'))$, but by our contrary assumption we have $g(\phi(v)) = g(\phi(f^{-1}(t)))$, which implies $\phi(v) = \phi(f^{-1}(t))$, which implies $v \in \Phi_{f^{-1}(t)}$. Hence, we reached a contradiction since any edge (v, v') with $v \in \Phi_{f^{-1}(t)}$ is removed in the definition of $\widetilde{F}_t$.

Hence, when $(v, v') \in \widetilde{F}_t$, we can assume $g(\phi(v)), g(\phi(f^{-1}(t))), g(\phi(v'))$ are all distinct. Hence, the property of $g(\phi(v)) \leqslant g(\phi(f^{-1}(t))) \leqslant g(\phi(v'))$ of the numbering in Lemma 3.5b can be strengthened to $g(\phi(v)) < g(\phi(f^{-1}(t))) < g(\phi(v'))$. By the construction of Γ' there is a path

$$(\phi(v) =: w_0, w_1, w_2, \cdots, w_m := \phi(v'))$$

of length not more than 2κ with endpoints $\phi(v)$ and $\phi(v')$ in Γ'. Then it is 'obvious' there is an edge in the path (w_i, w_{i+1}) in the path such that $g(w_i) \leqslant g(\phi(f^{-1}(t))) < g(w_{i+1})$. It follows that $(w_i, w_{i+1}) \in G_t$ and it corresponds to (v, v') in case 2. Hence, the correspondence presented satisfied the requirements, and the inequality is shown.

A note on why is this 'obvious': assume the contrary that there is no such edge, number each vertex, and compare it with $g(\phi(f^{-1}(t)))$. Intuitively, each vertex comes with a tag of " $<$ ", " $=$ ", " $>$ ", with the leftmost tag must be " $<$ ", rightmost tag must be " $>$ ", and at most one " $=$ ". If there is no equal, we can always find a " $<$ ", " $>$ " in a neighbouring pair by "squeezing", or else, we can still find a " $=$ ", " $>$ ", which this edge is also in G_t. □

Theorem 3.9. Let X, Y be bounded degree graphs. If there is a regular map $\phi : VX \to VY$, then,

$$\mathrm{cw}_X \lesssim \mathrm{cw}_Y.$$

Proof. By Proposition 3.8, going through all t, for all numbering g on Γ' there exists numbering f on Γ and $1 \leqslant k \leqslant |\Gamma|$ such that

$$\begin{aligned}\max_t \#\{v, v' \in E\Gamma : f(v) \leqslant t < f'(v)\} &\leqslant C \cdot \#\{w, w' \in E\Gamma' : g(w) \leqslant g(\phi(f^{-1}(k))) < g(w')\} \\ &\leqslant \max_\ell C \cdot \#\{w, w' \in E\Gamma' : g(w) \leqslant \ell < g(w')\}\end{aligned}$$

At right hand side of the equation, there is a minimum for which will be obtained by some g, so there is a numbering f on Γ such that:

$$\max_t \#\{v, v' \in E\Gamma : f(v) \leqslant t < f'(v)\} \leqslant \min_g \max_\ell C \cdot \#\{w, w' \in E\Gamma' : g(w) \leqslant \ell < g(w')\}$$

$$\min_f \max_t \#\{v, v' \in E\Gamma : f(v) \leqslant t < f'(v)\} \leqslant \min_g \max_\ell C \cdot \#\{w, w' \in E\Gamma' : g(w) \leqslant \ell < g(w')\}$$

So $\mathrm{cw}(\Gamma) \leqslant C\mathrm{cw}(\Gamma')$. □

3.4. **Monotonicity of Treewidth.**

Lemma 3.10. If Γ is a finite graph, For any subdivision $\Gamma \mapsto \gamma$, we have $\mathrm{tw}(\Gamma) \leqslant 2\mathrm{tw}(\gamma)$.

Proof. Given a subdivision $\Gamma \mapsto \gamma$, and given a tree subdivision (x_i, T) of γ, construct a subdivision (X_i, T) on Γ as follows:

- Every node vertices x_i can be decomposed into a disjoint union $x_i = (x_i \cap V\Gamma) \cup (x_i \setminus V\Gamma)$.
- If $w \in x_i \setminus V\Gamma$, then it is a subdivision vertex, which means it is added to γ from an edge $(u, v) \in E\Gamma$.
- Now define

$$X_i = (x_i \cap V\Gamma) \cup \bigcup_{w \in x_i \setminus V\Gamma} \{u, v : w \text{ is added from subdividing the edge } (u, v) \in E\Gamma.\}.$$

Now it suffices to check (X_i, T) is a tree decomposition, and notice that obviously $|X_i| \leqslant 2|x_i|$ for all $i \in I$ $(= VT)$, hence for any tree decomposition in γ with width w, there is a tree decomposition of Γ with width not more than $2w$, hence the inequality follows. Now we check the condition for tree decomposition:

- Condition 1.6a: Since (x_i, T) is a tree decomposition of γ: we have $\bigcup_{i \in I} x_i = V\gamma$. Hence $\bigcup_{i \in I} (x_i \cap V\Gamma) \cup (x_i \setminus V\Gamma) = V\gamma$. This implies $\bigcup_{i \in I} (x_i \cap V\Gamma) = V\Gamma$ since the union (inside) is disjoint. Also, for each i, $(x_i \cap V\Gamma) \subseteq X_i$ by the definition of X_i. Hence $\bigcup_{i \in I} X_i = V\Gamma$ as required.
- Condition 1.6b: Consider an edge $(u, v) \in E\Gamma$, there are two cases:

Case 1. If $(u, v) \in E\gamma$, it means the edge has not been subdivided from $\Gamma \mapsto \gamma$. Then by (x_i, T) being a tree decomposition, we can find x_i such that $u \in x_i, v \in x_i$. But since $u \in V\Gamma, v \in V\Gamma$, this would mean $u \in X_i, v \in X_i$, as required.

Case 2. Otherwise, the edge $(u, v) \in E\Gamma$ has been subdivided into a path $(u, w, \cdots v)$. Where $\cdots$ represent other possible subdivision vertices. Now consider $(u, w) \in V\gamma$, since (x_i, T) is a tree decomposition, there is x_i such that $u \in x_i, w \in x_i$. Now by the definition of X_i, since $u \in V\Gamma$, and w is a subdivision vertex between (u, v), it follows that $u \in X_i, v \in X_i$ as required.

- Condition 1.6c: Consider $i \in I, k \in I$ such that $v \in X_i \cap X_k$. Let $j \in I$ be a vertex in the unique path containing i and k. There are three cases:

Case 1. $v \in x_i \cap x_k$. In this case, since (x_i, T) is a tree decomposition, every node vertices x_j contains v. Then by the definition of X_j and $v \in V\Gamma$, it follows that every node vertices X_j contains v as well.

Case 2. $v \notin x_i \cap x_k$, and $v \in x_k$. Since $v \in X_i$. it follows that there is $w_1 \in x_i$, $u_1 \in \Gamma$ such that w_1 is a subdivision vertices from an edge $(v, u_1) \in E\Gamma$. Now consider the path

$$(u_1, \cdots w_1, w_1^1, w_2^1, \cdots, w_{m-1}^1, v)$$

on γ. Then we inductively use the fact that (x_i, T) is a tree decomposition of γ as follows:

- Since $(w_1, w_1^1) \in E\gamma$, there is node vertex x_{i_1} such that $w_1 \in x_{i_1}, w_1^1 \in x_{i_1}$.
- Since $w_1 \in x_i$, $w_1 \in x_{i_1}$, the unique path (in T) connecting i and i_1 contains w_1 in all the node vertices.
- Since $(w_1^1, w_2^1) \in E\gamma$, there is node vertex x_{i_2} such that $w_1^1 \in x_{i_2}, w_2^1 \in x_{i_2}$.
- Since $w_1^1 \in x_{i_1}, w_1^1 \in x_{i_2}$, the unique path connecting i_1 and i_2 contains w_1^1 in all the node vertices.
- Continue by considering $(w_2^1, w_3^1) \in E\gamma$...
- $\cdots$
- Since $(w_{m-1}^1, v) \in E\gamma$, there is node vertex x_{i_m} such that $w_{m-1}^1 \in x_{i_m}, v \in x_{i_m}$.

* Since $w^1_{m-1} \in x_{i_{m-1}}.w^1_{m-1} \in x_{i_m}$, the unique path connecting i_{m-1} and i_m contains w^1_{m-1} in all the node vertices.
* Since $v \in x_{i_m}, v \in x_k$, the unique path connecting i_m and k contains v in all the node vertices.

Combining, we obtain a sequence of vertices in T (i.e. the index I):

$$\mathcal{Q} := (i --- i_1 --- i_2 \cdots --- i_m --- k)$$

where

* $i_1 --- i_2$ signifies all the vertices in T in the unique path connecting i_1 and i_2,
* Each neighbouring vertex in $\mathcal{Q}$ is an edge in T, and
* All node vertices x_q such that $q \in \mathcal{Q}$ contains at least one of the following: w_1, w^1_j for some $1 \leqslant j \leqslant m-1$, or v. Hence, by the construction of (X_i, T), X_q will contain v for all $q \in \mathcal{Q}$.

Since T is a tree, there is a unique path connecting i, k. It follows that $\mathcal{Q}$ contains all vertices j of that unique path. Then by the last point above we have $v \in X_j$ for all j, which is as required.

Case 3. $v \notin x_i$ and $v \notin x_k$. Since $v \in X_i \cap X_k$. it follows that there is $w_1 \in x_i, w_2 \in x_k, u_1 \in V\Gamma, u_2 \in V\Gamma$ such that

* w_1 is a subdivision vertex from an edge $(v, u_1) \in E\Gamma$.
* w_2 is a subdivision vertex from an edge $(v, u_2) \in E\Gamma$.

Now, consider the path:

$$(u_1, \cdots w_1, w^1_1, w^1_2, \cdots, w^1_{m-1}, v, w^2_{n-1}, w^2_{n-2} \cdots w^2_2, w^2_1, w_2 \cdots u_2).$$

Now consider $\gamma_1 := (u_1 \cdots w_1, w^1_1, w^1_2, \cdots, w^1_{m-1}, v)$ and $\gamma_2 = (u_2 \cdots w_2, w^2_1, w^2_2, \cdots, w^2_{n-1}, v)$ separately, and handle them with the same procedure as the above case, except since $v \notin x_k$, we cannot carry out the last step. But we can still obtain two sequences of vertices in T, namely (since $w_1 \in x_i, w_2 \in x_k$):

$$\mathcal{Q}_1 := (i --- i_1 --- i_2 \cdots --- i_{m-1} --- i_m)$$
$$\mathcal{Q}_2 := (k --- j_1 --- j_2 \cdots --- j_{n-1} --- j_n)$$

That is, for the last step of the inductive process, we have:

* For γ_1: Since $(w^1_{m-1}, v) \in E\gamma$, there is node vertex x_{i_m} such that $w^1_{m-1} \in x_{i_m}, v \in x_{i_m}$.
* Since $w^1_{m-1} \in x_{i_{m-1}}.w^1_{m-1} \in x_{i_m}$, the unique path connecting i_{m-1} and i_m contains w^1_{m-1} in all the node vertices.
* For γ_2: Since $(w^2_{n-1}, v) \in E\gamma$, there is node vertex x_{j_n} such that $w^2_{n-1} \in x_{j_n}, v \in x_{j_n}$.
* Since $w^2_{n-1} \in x_{j_{n-1}}.w^2_{n-1} \in x_{j_n}$, the unique path connecting j_{n-1} and j_n contains w^2_{n-1} in all the node vertices.

Now, since $v \in x_{i_m}, v \in x_{j_n}$, the unique path connecting i_m and j_n contains v in all the node vertices. That means the sequence $\mathcal{Q}_1, \mathcal{Q}_2$ can be concatenate at the end:

$$\widetilde{\mathcal{Q}} = (i --- i_1 --- i_2 \cdots --- i_{m-1} --- i_m --- j_n --- j_{n-1} \cdots --- j_1 --- k).$$

For which, it also satisfies the condition similar to the last case, namely:

* Each neighbouring vertex in $\mathcal{Q}$ is an edge in T, and
* All node vertices x_q such that $q \in \widetilde{\mathcal{Q}}$ contains at least one of the following: w_1, w^1_ℓ, w^2_t for some $1 \leqslant \ell \leqslant m-1, 1 \leqslant t \leqslant n-1$, or v.

Hence, by the construction of (X_i, T), X_q will contain v for all $q \in \widetilde{\mathcal{Q}}$. Then we also have all j that is in the path of i and k, $v \in x_j$ as required.

□

Lemma 3.11. Let X, Y be graphs, if $f : VX \to VY$ is a simplicial map, then $\mathrm{tw}(X) \leqslant (\max_{w \in VY} |f^{-1}(w)|) \cdot \mathrm{tw}(Y)$.

Proof. Any tree composition of Y corresponds to a tree composition of X as follows: given (Y_i, T) as a tree decomposition of Y, define a tree decomposition of X by taking the same tree T, and define $X_i = f^{-1}(Y_i)$. Again, as in Lemma 3.10, clearly $|X_i| \leqslant (\max_{w \in VY} |f^{-1}(w)|) \cdot |Y_i|$, hence the required inequality will follow. Now we check the conditions, which is routine.

- Condition 1.6a: Since (Y_i, T) is a tree decomposition. Given $x \in VX$, there is Y_i such that $f(x) \in Y_i$, so for all x we have $x \in X_i$. Hence $\bigcup_{i \in I} X_i = VX$.
- Condition 1.6b: Let (u, v) be an edge in EX. Then since f is simplicial, either $f(u) = f(v)$, for which by 1.6a on (Y_i, T), we have $f(u) = f(v) \in Y_i$ for some $i \in T$, then it follows $u \in X_i$ and $v \in X_i$; or $(f(u), f(v)) \in EY$, for which by 1.6b on (Y_i, T), there is Y_j, $j \in I$ such that $f(u) \in Y_j$, $f(v) \in Y_j$, which means $u \in X_j, v \in X_j$ as required.
- Condition 1.6c: Let $i, j, k \in VT$. Notice that for any set, $Y_i \cap Y_k \subset Y_j$ will imply $f^{-1}(Y_i) \cap f^{-1}(Y_k) \subset f^{-1}(Y_j)$. So when j is in the unique path connecting i and k, by (Y_i, T) is a tree decomposition we have $Y_i \cap Y_k \subseteq Y_j$, and that directly implies $X_i \cap X_k \subseteq X_j$ as required.

□

Theorem 3.12. Let X, Y be bounded degree graphs. If there is a regular map $\phi : VX \to VY$, then,

$$\mathrm{tw}_X \lesssim \mathrm{tw}_Y.$$

Proof. By Proposition 3.4, given any finite subgraph $\Gamma \leqslant X$, write $\phi = g \circ f$, where g is a simplicial map.

First we bound the quantity $\max_{v \in V\Gamma'} |g^{-1}(w)|$, we investigate the algorithm in constructing the function g in Proposition 3.4. Fixing $w \in V\Gamma'$. If $g^{-1}(w) \subseteq V\Gamma$ then since g only send element in $V\Gamma$ according to ϕ, in this case $|g^{-1}(w)| = |\phi^{-1}(w)| \leqslant \kappa$.

Else, $g^{-1}(w) \not\subseteq V\Gamma$. That means $w \in V\Gamma'$ is in at least one of the path $\mathcal{P}_{(v,v')}$ for some $\{\Phi_v, \Phi_{v'}\} \in \mathcal{S}$. Since ϕ is κ-regular, there are at most κ vertices in Φ_v and $\Phi_{v'}$. So when repeating step 3 (in step 4), there are at most κ^2 edges chosen, all waiting to be subdivided from a fixed path $\mathcal{P}_{(v,v')}$ containing the vertex w as the interior point. Now by Lemma 3.2, there are at most $\kappa \deg(Y)^\kappa$ paths passing through w in Γ'. Combining, we have $|g^{-1}(w)| \leqslant \kappa^3(\deg(Y))^\kappa$ for all $w \in V\Gamma$.

Now, combining Lemma 3.10 and 3.11, then for all $\Gamma \leqslant X$, there is Γ' such that $\mathrm{tw}(\Gamma) \leqslant 2 \max_{v \in V\Gamma'} |g^{-1}(w)| \cdot \mathrm{tw}(\Gamma')$. So

$$\mathrm{tw}(\Gamma) \leqslant 2\kappa^3(\deg(Y))^\kappa \cdot \mathrm{tw}(\Gamma') \tag{14}$$

$$\mathrm{tw}(\Gamma) \leqslant \max\{2\kappa^3(\deg(Y))^\kappa \mathrm{bw}(\Gamma') : \Gamma' \leqslant Y, |\Gamma'| \leqslant 2\kappa \deg(X)|\Gamma|\} \qquad \text{by Lemma 3.1.}$$

$$\mathrm{bw}(\Gamma) \leqslant 2\kappa^3(\deg(Y))^\kappa \mathrm{tw}_Y(2\kappa \deg(X)|\Gamma|) \tag{15}$$

now for arbitrary Γ such that $|\Gamma| \leqslant r$, (15) holds, it follows that $\mathrm{tw}_X(r) \lesssim \mathrm{tw}_Y(r)$ as required. □

4. Conclusion and Further Questions

In Proposition 2.7, we see that $\mathrm{sep}_G \lesssim \mathrm{tw}_G$, it is not known whether indeed $\mathrm{sep}_G \simeq \mathrm{tw}_G$.

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